

spiselectscw
Explanatory Guide and Users Manual

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Chapter 1

Introduction

The aim of *spiselectscw* is to select science windows from the ScW database, according to criteria given by the user, and then produce a new database with the selected ScW, and, optionally, create files (fits and ASCII) compatible with *spiaddobs* and the standard processing of Jurgen Knodlseder's softwares (*spi_obs_add*). This software complies with the ISDC conventions (especially for the parameters file). Moreover, it can also build the corresponding observation group from preprocessed data and create an 'simplified' exposure map.

spiselectscw can take a selection of the "basic" science window database files as arguments (which should then comply with the repository architecture described above) and can also make use of a user supplied database (e.g. the result of *spiselectscw*).

The selection criteria of ScWs are defined in the parameters file (*spiselectscw.par*). Some of these criteria are pre-defined (such as selection on revolution, pointings, time, etc), other 'ranges' or 'value' criteria can be defined for any elements of any columns of the fits file(s). Moreover, a more general expression, conforming to the fitsio selection syntax can be given to the program. Any of these criteria can be omitted (or turned into comments) in the parameters file, in which case no check is done on this criterion, nor is checked the validity of the values. As a side effect, if one only wants to discard NULL values from a column, the user should use a selection criterion which will always be satisfied for that column.

spiselectscw is written in ANSI C++ language and was developed under ISDC platforms OSA 3.0 to 5.2.

Chapter 2

Getting started

Either you are using a binary version of *spiselectscw* or want compile it from sources, the OSA environment has to be set. At MPE, this is done by sourcing the configuration script locating in the OSA subdirectory :

```
# source <Path_to_osa>/init_osa best
```

<Path_to_osa> stands for `/afs/ipp-garching.mpg.de/mpe/gamma/instruments/integral/isdc/osa` at the MPE. In the above command line, `#` stands for the UNIX or LINUX prompt (don't type it !).

Then, you have to set the `PFILES` environment variable to the directory of the parameter file. Usually, the parameter file is in the working directory, and so you should type :

```
# setenv PFILES .
```

2.1 Precompiled binaries

At the MPE, the documentation, parameter file and pre-compiled binaries of additional SPI software are in subdirectories of `/afs/ipp/mpe/gamma/instruments/integral/software`. So, in *spiselectscw/4.0*, one can find :

- the program binaries in subdirectories whose name reflect the machine architecture and compiler. Their name is then of the form `SYSARCH_COMP`, where `SYSARCH` is the system architecture (as defined by the `SYS` environment variable, type `echo $SYS` to see its value), and `COMP` is the compiler (`gcc`, `g++`, `icc`, `icpc`, ...).
- the documentation directory (`doc`) that should, at least, contain a user manual called `spiselectscw.ps`.
- an example of parameter file, called `spiselectscw.par`.

If you are working on a machine for which a binary version exists, you probably have to use it. The next section (compiling from the source) is mostly intended for expert user in case of bug correction or compilation on a new system.

2.2 Compiling from sources

The source directory of additional SPI software can be found by following the `/afs/ipp-garching.mpg.de/mpe/gamma/instruments/integral/users/halloyin/C` symbolic link. In the `spiselectscw/4.0` subdirectory, the following items should be present :

- source (*.cpp, *.c) and header (*.h) C++ and/or C files
- an `ac_stuff` subdirectory containing a slightly modified (in order to support 64 bits machine) configuration scripts from ISDC.
- a `doc` containing the software documentation
- a `makeisdc.in` file, used to configure the installation process.

Assuming that all required libraries are available for the appropriate machine and compiler version, the compilation procedure is the following :

1. Clean any previous installation :

```
# make distclean
```

Please note that you may have to use `gmake` instead of `make` on some systems.

2. Configure the current installation :

```
# ./ac_stuff/configure
```

This configure script accepts many parameters and optional configuration flags. You may have an overlook of these option by typing `# ./ac_stuff/configure --help`. In particular, the compiler name and options can be set manually on the configure line by typing :

```
# ./ac_stuff/configure CC=... CFLAGS=... CXX=... CXXFLAGS=...,
```

where `CC` is the name of the C compiler and `CFLAGS` is the list of options to pas to the C compiler. `CXX` and `CXXFLAGS` follo the same definition for the C++ compiler. At the MPE, a compilation with the Intel compiler and a optimized code for Pentium 4 an compatible CPUs would then be :

```
# ./ac_stuff/configure CC=icc CFLAGS=-axW CXX=icpc CXXFLAGS=-axW,
```

The program will anyhow be compiled with the `-O3` and `-Wno-deprecated` options.

3. Build the binary :

```
# make all
```

This should compile the program and put it in a subdirectory called `SYS_CXX`, where `SYS` is the system architecture (as defined by the `SYS` environment variable, type `echo $SYS` to see its value), and `CXX` is the compiler (`gcc`, `g++`, `icc`, `icpc`, ...).

What to do if anything went wrong in one of the previous step ? Well ... First look carefully at the error message(s) to see if it is not an OSA or local system configuration problem. Second, if it seems to be a bug in any of the configuration script or even in the source code, you may consider to get one (or maybe much more ?) cup of coffee and dive into the code and scripts. In desperate cases, you may also contact the programmer to complain about the software (eventually, (s)he will be able to identify and maybe solve your problem...).

2.3 Software versions and supported platforms

2.3.1 Versionning information

Date	Version	Comments
?? May 2004	1.0	First release
10 Dec 2004	3.4	First documented version
28 Jul 2006	3.9	(Zhang X.L.) Modified for new version of OSA (handle dead detectors properly)
06 Nov 2012	4.0	Re-enable filtering of dead detectors, increase the maximum file size to 1.75 GB, solve minor bugs.

2.3.2 Supported platforms

The following table gives some information about the platforms on which *spiselectscw* has been compiled :

- System : value of the **SYS** environment variable
- Platform : hardware architecture, as given by **uname -i**
- Kernel version : as given by **uname -rs**
- Operating system : as given by **uname -o**

Vers.	OSA	system	platform	kernel	OS	compiler
1.0	5.1/linux	i386.rh90	i386	Linux 2.4.21-241-smp	GNU/Linux	g++/gcc icpc/icc

2.3.3 Known issues

- The DEADC keyword (global dead time correction) may be incorrect when filtering already-filtered data with dead detectors (the present computation assumes that the initial DEADC don't take dead detectors into account, which may be wrong ...

Chapter 3

General purpose parameters

The parameters listed here are for general purpose and do not influence the behavior of the program.

3.1 Debugging flag

Parameter	: debug
Type	: integer
Range	: ≥ 0
Description	: Increasing integer gives more verbose output. Can be used in case of error and for development purposes.

3.2 Information parameters

The values of the following parameters will be written in the fits files headers for user convenience and documentation.

3.2.1 observation group identifier

Parameter	: group_id
Type	: string
Range	:
Description	: Observation group identifier. Recorded for information in the output files headers.

3.2.2 observation identifier

Parameter	: observation_id
Type	: string
Range	:
Description	: Observation identifier. Another information string..

3.2.3 Scientific purpose

Parameter	: purpose
Type	: string
Range	:
Description	: More detailed info on the scientific purpose of the observation. Will be split over multiple lines if more than 80 characters and recorded in output headers.

3.2.4 Observed object

Parameter	: object
Type	: string
Range	:
Description	: Observed Object

3.2.5 Observer

Parameter	: observer
Type	: string
Range	:
Description	: The name of the observer

Chapter 4

Input and output files location

The program need some input data files (such as the ScW database, pre-processed observations, etc.) and creates output FITS and text files. Some of these files are optional (either in input or output). The following parameters are used to define their location and/or their existence.

4.1 File size limitation, split data files

spiselectscw makes use of the CFITSIO library. In general, CFITSIO has an internal limitation on the size of the file of 2.1 GBytes (2^{31} bytes, that's a fundamental limitation of some file access function of the standard C compilers using an int datatype as file pointers). When compiled with the proper compilation flags (see http://heasarc.nasa.gov/docs/software/fitsio/c/f_user/node13.html), this file limitation can be increased to about 6 TBytes on 64 bits machines.

As the configuration of the CFITSIO library is unknown, and to keep some margin, the size of the output files produced by spiselectscwis is limited to 1.75 GBytes. If a given file is about to exceed this limit, a new file is automatically created (using the same filename concatenated with an index number).

When a new file is created, a NEXTFILE keyword is appended to the 'old' file header indicating the name of the continuing file. Symmetrically a PREVFILE keyword is appended to the header of the new file, to indicate the name of the previous file. A SPLITNUM keyword is also added to specify the sequence number.

4.2 Location of the input files

4.2.1 Explicit name of a ScW database file

Parameter	: <code>fits_input_file</code>
Type	: string
Range	:
Description	: DOL (with HDU specification) of a user supplied ScW database, as created by <i>spimkdbase</i> . Can be omitted if a revolution list is provided (see <code>fits_revolutions_list</code>).

4.2.2 List of revolutions to process

Parameter	: <code>fits_revolutions_list</code>
Type	: string
Range	: ≥ 0
Description	: Comma separated list of revolutions to process. '-' indicates an interval of revolutions (e.g : "10-20,35" stands for revolutions 10,11,12,...,20,35). This parameter is optional if the user supplies directly a ScW database (see <code>fits_input_file</code>). The archive directory and the prefix for filenames are defined through <code>fits_rev_data_dir</code> and <code>fits_rev_data_pref</code> .

4.2.3 Prefix for the pre-processed ScW database file names

Parameter	: <code>fits_rev_data_pref</code>
Type	: string
Range	:
Description	: Common prefix for the ScW parameters files which is formed with the format : <code><fits_rev_data_pref>RRRR.fits</code> , where RRRR is the revolution number

4.2.4 Directory of the pre-processed ScW database file

Parameter	: <code>fits_rev_data_dir</code>
Type	: string
Range	:
Description	: Relative or absolute path to the science window database archives, i.e the directories <code><fits_rev_data_dir>/RRRR/</code> (where RRRR is the revolution number) should contain the fits ScW parameters files (see <code>fits_rev_data_pref</code>). This parameter is mandatory if a revolution list is specified, ignored otherwise.

4.2.5 Prefix for the pre-processed data location

Parameter	: <code>rev_std_proc_pref</code>
Type	: string
Range	:
Description	: Common prefix for the location of the standard analysis OG. The location of the preprocessed observations is expected to be <code><rev_std_proc_pref>RRRR<rev_std_proc_suff>/<rev_std_og_file></code> , where RRRR is the revolution number.

4.2.6 Suffix for the pre-processed data location

Parameter	: <code>rev_std_proc_suff</code>
Type	: string
Range	:
Description	: Common suffix for the location of the standard analysis OG. The location of the preprocessed observations is expected to be <code><rev_std_proc_pref>RRRR<rev_std_proc_suff>/<rev_std_og_file></code> , where RRRR is the revolution number.

4.2.7 Name of the observation group file of the pre-processed data

Parameter	: <code>rev_std_proc_suff</code>
Type	: string
Range	:
Description	: Common name of the preprocessed OG fits file. The location of the pre-processed observations is expected to be <code><rev_std_proc_pref>RRRR<rev_std_proc_suff>/<rev_std_log_file></code> , where RRRR is the revolution number.

4.2.8 Name of the ScW group index for the pre-processed data

Parameter	: <code>rev_std_swg_idx</code>
Type	: string
Range	:
Description	: Name of the ScW group index in the preprocessed data.

4.2.9 Definition of the “Off-line” detectors

Parameter	: <code>off_line_detectors_file</code>
Type	: string
Range	:
Description	: Name of the fits file containing the list of the off-line detectors (none if left empty)

Parameter	: <code>off_line_dead_ratio</code>
Type	: real
Range	:
Description	: The value to be put in the DEAD_RATIO column for the dead detectors (as read from the off-line detectors file). No update of the LIVETIME, COUNTS, STAT_ERR and DEAD_RATIO columns if set to a negative value.

4.3 Input/Output mode for ScW database

The program can create a new ScW database file and build the observation from it or directly use a already processed ScW database file.

Parameter	: <code>create_scw_file</code>
Type	: int
Range	: 0,1
Description	: Create (1) or read (0) from the specified ScW file

Parameter	: <code>scw_file</code>
Type	: string
Range	:
Description	: Name of the input/output selected ScWs database

4.4 File names of the output files

4.4.1 List of selected ScWs

Parameter	: <code>scwlist_output_file</code>
Type	: string
Range	:
Description	: Path and name of the ScW list, i.e. an ASCII file compatible with the standard processing of Jurgen Knodlseder's softwares (<i>spi_obs_add</i>). If this parameter is omitted, no ScW list is produced. See also <code>scw_base_dir</code> for the specification of the ScW directory.

n

4.4.2 List of observation groups

Parameter	: <code>oglist_output_file</code>
Type	: string
Range	:
Description	: Path and name of the 'Observation Group' fits file, for use with <i>spi_addobs</i> . This fits file is actually an index file containing the DOLs of the <code>og_spi.fits</code> index files corresponding to the standard revolution processing. If this parameter is left blank, no OG list is produced. See also <code>rev_std_proc_pref</code> , <code>rev_std_proc_suff</code> for the specification of the standard revolution processing directory, and <code>rev_frac_sel</code> for the selection criterion of one revolution.

4.4.3 Name of the created observation group

Parameter	: group_output_file
Type	: string
Range	:
Description	: Name of the output group, None if left blank

4.4.4 Name of the energy boundaries file

Parameter	: ebounds_output_file
Type	: string
Range	:
Description	: Name of the output ebounds file, mandatory if group_output_file not blank.

4.4.5 Name of the pointings definition file

Parameter	: pointing_output_file
Type	: string
Range	:
Description	: Name of the output pointings file, mandatory if group_output_file not blank.

4.4.6 Name of the GTI definition file

Parameter	: gti_output_file
Type	: string
Range	:
Description	: Name of the output gti file, mandatory if group_output_file not blank.

4.4.7 Name of the detector events file

Parameter	: dspec_output_file
Type	: string
Range	:
Description	: Name of the output detector events file, mandatory if group_output_file not blank.

4.4.8 Name of the dead time file

Parameter	: <code>dtime_output_file</code>
Type	: string
Range	:
Description	: Name of the output dead time file, mandatory if <code>group_output_file</code> not blank.

4.4.9 Creation of a ScW index file

Parameter	: <code>attach_swg_idx</code>
Type	: int
Range	: 0-1
Description	: Attach SwG index to the output OG (if created)

Parameter	: <code>out_swg_idx_dol</code>
Type	: string
Range	:
Description	: Name of the created ScW group index file. Non created if left blank.

4.4.10 Exposure map location

Parameter	: <code>out_expo_map_dol</code>
Type	: string
Range	:
Description	: Name of the output exposure map. None created if left blank

4.5 Exposure map parameters

4.5.1 Exposure map building method

Parameter	: <code>expo_map_from_0G</code>
Type	: int
Range	: 0-1
Description	: The exposure map can be calculated from the pointing data of the created observation (if 1) or from the information in the ScW database (if 0). The latter method is faster but less precise.

4.5.2 Coordinate system and projection method

Parameter	: expo_map_coordsys
Type	: string
Range	: Galactic/Ecliptic/J2000/B1950
Description	: Coordinate system of the exposure map

Parameter	: expo_map_proj
Type	: string
Range	:
Description	: Projection method of the exposure map. Should correspond to one of the projection methods of the cfitsio library.

4.5.3 Longitude range and binning

Parameter	: expo_map_chi_min
Type	: real
Range	:
Description	: Longitude in degrees of the first “left” pixel of the map

Parameter	: expo_map_chi_max
Type	: real
Range	:
Description	: Longitude in degrees of the last “right” pixel of the map

Parameter	: expo_map_chi_delta
Type	: real
Range	:
Description	: Longitude size of pixels in degrees.

4.5.4 Latitude range and binning

Parameter	: expo_map_psi_min
Type	: real
Range	:
Description	: Latitude in degrees of the first “left” pixel of the map

Parameter	: expo_map_psi_max
Type	: real
Range	:
Description	: Latitude in degrees of the last “right” pixel of the map

Parameter	: expo_map_psi_delta
Type	: real
Range	:
Description	: Latitude size of pixels in degrees.

Chapter 5

Selecting ScWs and building the observation

The first work of the program is to select ScWs according to a set of user-defined parameters. Then, an observation group (i.e. containing all the required observation files) can be (optionally) created. So, apart from the selection on ScWs parameters, the final observation can also be filtered according to, e.g., time ranges, detector and energy selection.

5.1 Foreword on selection lists

For many of the selection parameters described below, the expected criterion is a list of intervals and/or single values.

The program allows to define many ranges and/or values at once by entering a comma separated list of real or integer ranges or values. For example, the list of reals ‘‘1073.45-1142, 1145-1303.5, 1304.7-1416’’ defines three real intervals. Concerning integer selection, a list like ‘‘0-10, 15, 20-30, 35’’ would select integers from 0 to 10, 15, 20 to 30 and 35.

Unless specified, a given selection is not performed if no argument is provided.

5.2 Selection on the output observation

5.2.1 Time selection

To account for slight mismatches between pointings and ScW boundaries, a time tolerance has to be set on the ScW boundaries. It is set by :

Parameter	: <code>time_tolerance</code>
Type	: real
Range	: > 0
Description	: Time tolerance on ScW time limits in seconds

Parameter	: <code>telapse_selection</code>
Type	: list of real intervals
Range	:
Description	: List of accepted elapsed time per pointing (in s)

Parameter	: <code>livetime_selection</code>
Type	: list of real intervals
Range	:
Description	: List of accepted livetime per pointing (summed over detectors selection) (in s)

Parameter	: <code>det_livetime_min</code>
Type	: real
Range	:
Description	: Minimum accepted livetime per detector (useful to reject dead detectors)

5.2.2 energy selection and (re)binning

Parameter	: <code>energy_bins</code>
Type	: list of real intervals
Range	:
Description	: List of selected energy ranges. If 'keV' is placed at the end of the list, then the ranges are assumed to be expressed in keV. Otherwise, they are assumed to refer to input energy bins number (beginning at 1).

Parameter	: <code>energy_bins</code>
Type	: vector of real numbers
Range	:
Description	: List of energy (re)binning per output energy range. If the number of values is less than the defined energy ranges, the last binning value is used for the remaining energy ranges. If 'keV' is placed at the end of the list, then the values are assumed to be expressed in keV. Otherwise, they are assumed to refer to a number of input energy bins per output bin.

Parameter	: <code>collect_energies</code>
Type	: int
Range	: 0-1
Description	: If set to 1, then the different energy bins (and corresponding events) are collected into 1 single bin. In that case, the true “size” of the energy range is lost (i.e. the output file record only the global min and max boundaries).

5.2.3 Detector selection

Parameter	: <code>det_selection</code>
Type	: list of integers ranges and values
Range	:
Description	: List of accepted detectors in the output observation.

5.3 Output OG list parameters

Parameter	: <code>rev_frac_sel</code>
Type	: float
Range	: ≥ 0
Description	: Since the selection is made on ScW and the OG list contains only entry per revolution, it is necessary to define a criterion for the selection of a revolution. <i>spiselectscw</i> uses a exposure threshold for that purpose, controlled with <code>rev_frac_sel</code> : if the exposure time (*NOT* the live time) of the selected ScW represents a fraction greater than <code>rev_frac_sel</code> of the exposure time read in the header of the (<code>og_spi.fits</code>) file of a given revolution, then this revolution is selected and added to the OG list. Please note that because of the filtering of ScW of the standard revolution processing, its elapsed time is lower than the total exposure time available in ScW from this revolutions and so, the exposure time of selected ScW can be greater than the one used for revolutions processing. More generally, the ScW selection criteria should contain the revolution processing criteria for the <code>rev_frac_sel</code> parameter to be relevant. This parameter is mandatory if an OG list is created, ignored otherwise.

5.4 Output ScW list parameters

Parameter	: <code>scw_base_dir</code>
Type	: string
Range	:
Description	: Path (usually relative to <code>REP_BASE_PROD</code>) of the ScW archive, for use with <i>spi_obs_add</i> . This parameter is mandatory if a ScW list is specified, ignored otherwise.

5.5 Predefined selection parameters

Parameter	: <code>revolutions_cond_value</code>
Type	: string
Range	:
Description	: Comma separated list of selected revolutions (<code>Revolution</code> column). '-' indicates an interval of revolutions (e.g : "10-20,35" stands for revolutions 10,11,12,...,20,35). This parameter is optional.

Parameter	: <code>ISOCptgs_cond_value</code>
Type	: string
Range	:
Description	: Comma separated list of selected ISOC pointing number (<code>ISOC_Pointing</code> column). '-' indicates an interval (e.g : "10-20,35" stands for 10,11,12,...,20,35). This parameter is optional.

Parameter	: <code>scwtype_cond_value</code>
Type	: string
Range	:
Description	: Comma separated list of selected science window type (<code>ScwType</code> column). '-' indicates an interval (e.g : "10-20,35" stands for 10,11,12,...,20,35). This parameter is optional.

Parameter : OBTstart_cond_value
Type : string
Range :
Description : Comma separated list of OBT ranges for the start of the ScW (OBTStart column). '-' indicates the interval and OBT can be given with a single value (8-bytes integer) or with a set of 4 values (4 2-bytes integer) (e.g : "0 2700 0 0 - 0 2900 0 0, 0 3000 0 0 - 0 3200 0 0" , equivalent to " 11596411699200 - 12455405158400 , 12884901888000 - 13743895347200"). This parameter is optional.

Parameter : OBTend_cond_value
Type : string
Range :
Description : Comma separated list of OBT ranges for the end of the ScW (OBTEnd column). '-' indicates the interval and OBT can be given with a single value (8-bytes integer) or with a set of 4 values (4 2-bytes integer) (e.g : "0 2700 0 0 - 0 2900 0 0, 0 3000 0 0 - 0 3200 0 0" , equivalent to " 11596411699200 - 12455405158400 , 12884901888000 - 13743895347200"). This parameter is optional.

Parameter : datestart_cond_value
Type : string
Range :
Description : Comma separated list of date ranges for the start of the ScW (DateStart column). '-' indicates the interval and dates should be given in one of the format YYYY-MM-DDThh:mm:ss or YYYY-MM-DD or dd/mm/yy or hh:mm:ss, **enclosed in " ' ' " characters**. If not mentioned hours, minutes and seconds will be set to 0. Conversely, if the string does not include year, month and day, then they are set to the corresponding values of the ScW being read (allows the selection of a part of each day), e.g. "'2003-01-01T00:00:00' - '2003-12-31T23:59:59'". This parameter is optional.

Parameter	: dateend_cond_value
Type	: string
Range	:
Description	: Comma separated list of date ranges for the end of the ScW (DateEnd column). '-' indicates the interval and dates should be given in one of the format YYYY-MM-DDThh:mm:ss or YYYY-MM-DD or dd/mm/yy or hh:mm:ss, <i>*enclosed in " " characters*</i> . If not mentioned hours, minutes and seconds will be set to 0. Conversely, if the string does not include year, month and day, then they are set to the corresponding values of the ScW being read (allows the selection of a part of each day), e.g. "2003-01-01T00:00:00" - "2003-12-31T23:59:59". This parameter is optional.

Parameter	: tstart_cond_value
Type	: string
Range	:
Description	: Comma separated list of ISDC Julian days ranges for the start of the ScW (TStart column). '-' indicates the interval and days are counted from January 1 st , 2000 at O ^h TT, e.g. "1150.5-1155.5". This parameter is optional.

Parameter	: tend_cond_value
Type	: string
Range	:
Description	: Comma separated list of ISDC Julian days ranges for the end of the ScW (TEnd column). '-' indicates the interval and days are counted from January 1 st , 2000 at 0 ^h TT (e.g. "1150.5-1155.5"). This parameter is optional.

5.6 Selection on observation type and ownership

5.6.1 Observation type (GPS, GCDE, etc)

Parameter	: obstype_cond_value
Type	: comma separated list of strings
Range	:
Description	: List of selected observation types (ObsType column), example "Calibration, GPS, GCDE". The observation type is set in the database according to information provided by ISDC.

5.6.2 Ownership

Parameter	: owner_cond_value
Type	: comma separated list of strings
Range	:
Description	: List of selected PI names (PIname column), combined with the proprietary status (Prop_Status column, usually “public” or “private”). The format of the list is “PIname1 – Status1, PIname2 – Status2, ...”. All ScWs for which one of the PI names contains “PIname1” (case insensitive) and its status is “Status1” will be selected. The same rule applies for “PIname2 - Status2”, etc. Wildcards characters (*,?,...) are accepted (* stands for any string and ? for any single character). The indication of the status is optional and replaced by * if omitted. So, “PIname1, ...” is equivalent to “PIname1 - *, ...”, meaning that all ScWs matching the PI name will be selected, regardless of their status. Alternatively, “* - public” will select all public ScWs, whatever the PI name.

5.7 Selection on arbitrary elements of logical columns

Parameter	: logical_cond_num
Type	: integer
Range	: ≥ 0
Description	: Number of logical columns to process. This parameter is optional.

Parameter	: logical_cond_colname_#l
Type	: string
Range	:
Description	: Name of the #l th logical column to process. This parameter is mandatory if, at least, #l logical columns should be processed (see logical_cond_num).

Parameter	: logical_cond_element_#l
Type	: integer
Range	: ≥ 0
Description	: Element of the #l th logical column to process, beginning at 1. Set to 1 if this parameter is not given (suitable for single columns).

Parameter	: <code>logical_cond_value_#l</code>
Type	: integer
Range	: 0,1
Description	: Selection criterion for the $\#l^{th}$ logical column to process, 0 = FALSE, 1 = TRUE. This parameter is mandatory if, at least, $\#l$ logical columns should be processed (see <code>logical_cond_num</code>).

5.8 Selection on arbitrary elements of numerical columns

Parameter	: <code>numerical_cond_num</code>
Type	: integer
Range	: ≥ 0
Description	: Number of numerical (float or integer) columns to process. This parameter is optional.

Parameter	: <code>numerical_cond_colname_#l</code>
Type	: string
Range	:
Description	: Name of the $\#l^{th}$ numerical column to process. This parameter is mandatory if, at least, $\#l$ numerical columns should be processed (see <code>numerical_cond_num</code>).

Parameter	: <code>numerical_cond_element_#l</code>
Type	: integer
Range	: ≥ 0
Description	: Element of the $\#l^{th}$ numerical column to process, beginning at 1. Set to 1 if this parameter is not given (suitable for single columns).

Parameter	: <code>numerical_cond_value_#l</code>
Type	: string
Range	:
Description	: Comma separated numerical ranges, '-' indicating the interval, e.g : "198.5 - 200.5, 210 - 215". This parameter is mandatory if, at least, $\#l$ numerical columns should be processed (see <code>numerical_cond_num</code>).

5.9 Selection on general expression

Parameter	: <code>select_GenrlSelExpr</code>
Type	: string
Range	:
Description	: Arbitrary general selection expression (but less than 8192 characters), conforming to the row filtering specification of the <code>cfitsio</code> , e.g “circle(0,0,10,L2_SCX,B2_SCX)” for selection of spacecraft pointings in a circle of 10 degrees centered on 0,0 in galactic coordinates. This parameter is optional.

5.10 Selection on spacecraft pointings

The S/C attitude is given in the ScW database for the two axes : X (pointing axis) and Z (perpendicular to X, towards the Sun). Besides these informations are available from the ScW group header (columns `RA_SCX`, `DEC_SCX`, `L2_SCX`, `B2_SCX`, `RA_SCZ`, `DEC_SCZ`, `L2_SCZ`, `B2_SCZ` : one value per ScW) and as a mean of the achieved pointing during the ScW from the ORBI-SCP HDU (columns `OrbitRA_SCX`, `OrbitDEC_SCX`, `OrbitL2_SCX`, `OrbitB2_SCX`, `OrbitRA_SCZ`, `OrbitDEC_SCZ`, `OrbitL2_SCZ`, `OrbitB2_SCZ`). The selection procedure is the same for the two axes and the different set of data, i.e. ‘masks’ (or sky region) are defined on the sky and the ScW is accepted/rejected if the mean pointing of the axis lies within one or more “masks”, depending on a logical equation. The models and parameters derived from the *gensskymodel* software are described in details in the section 6. Hereafter are a list of parameters needed for the masks’ definition.

Parameter	: <code>select_PtgX_masks_num</code>
Type	: integer
Range	: ≥ 0
Description	: Indicates the number of masks applied to the pointing selection of axis X according to the value of the ScW group header. This parameter is optional.

Parameter	: <code>select_PtgX_masks_globrev</code>
Type	: integer
Range	: 0, 1
Description	: If 1, globally reverses the effect of the masks applied to the pointing selection of axis X according to the value of the ScW group header. This parameter is mandatory if, at least, one mask is defined, ignored otherwise.

Parameter : `select_PtgX_masks_logeq`
Type : vector of (`select_PtgX_masks_num`-1) integers (string format)
Range : 0,1
Description : Logical equation specifying the operation with the next mask. 0 stands for OR, 1 for AND. Operations priorities are satisfied, i.e the contiguous AND operations are evaluated first and then the resulting blocks are combined with an OR operation. This parameter is mandatory if, at least, two masks are defined, ignored otherwise.

Parameter : `select_PtgX_masks_chi_list_#m`
Type : real
Range : -180 - +180 (recommended)
Description : χ (longitude) coordinate of the mask number `#m` for the pointing selection of axis X according to the value of the ScW group header, in the WCS given by (`select_PtgX_masks_coordsys_#m`). This parameter is mandatory if, at least, `#m` masks are defined, ignored otherwise.

Parameter : `select_PtgX_masks_chi_list_#m`
Type : real
Range : -180 - +180 (recommended)
Description : ψ (latitude) coordinate of the mask number `#m` for the pointing selection of axis X according to the value of the ScW group header, in the WCS given by (`select_PtgX_masks_coordsys_#m`). This parameter is mandatory if, at least, `#m` masks are defined, ignored otherwise.

Parameter : `select_PtgX_masks_angle_list_#m`
Type : real
Range :
Description : Rotation angle of the mask number `#m` for the pointing selection of axis X according to the value of the ScW group header. It is computed as the trigonometric angle (in degrees) between the great circle perpendicular to the local meridian (in the WCS of the mask) at the mask location and the equator of the mask model (i.e the great circle with $\psi_{model} = 0$). See the definition of the model for more details on its intrinsic coordinates system. This parameter is mandatory if, at least, `#m` masks are defined, ignored otherwise.

Parameter : `select_PtgX_masks_coordsys_#m`
Type : string
Range :
Description : Coordinate system of the mask number #m for the pointing selection of axis X according to the value of the ScW group header. B1950, J2000, GALACTIC and ECLIPTIC systems are currently supported. Only the indication of the first letter is required. This parameter is mandatory if, at least, #m masks are defined, ignored otherwise.

Parameter : `select_PtgX_masks_rev_#m`
Type : integer
Range : 0,1
Description : If 1, reverses the effect of the mask number #m for the pointing selection of axis X according to the value of the ScW group header.

Parameter : `select_PtgX_masks_name_#m`
Type : string
Range :
Description : Name of the mask model number #m for the pointing selection of axis X according to the value of the ScW group header. The value is case sensitive. See the description of the models for the valid model names. This parameter is mandatory if, at least, #m masks are defined, ignored otherwise.

Parameter : `select_PtgX_masks_npar_#m`
Type : integer
Range : ≥ 0 (model dependant)
Description : Number of parameters given model of the mask number #m for the pointing selection of axis X according to the value of the ScW group header. See the description of the model for the minimal number of parameters required and their signification. This parameter is mandatory if, at least, #m masks are defined, ignored otherwise.

Parameter	: <code>select_PtgX_masks_par_#m</code>
Type	: vector of <code>select_PtgX_masks_npar_#m</code> reals (string format)
Range	: model Dependant
Description	: Parameters given to the model of the mask number #m for the pointing selection of axis X according to the value of the ScW group header. See the description of the model for the valid values and their signification. This parameter is mandatory if, at least, #m masks are defined, ignored otherwise.

The same parameters apply for the pointing selection of axis Z according to the value of the ScW group header by replacing X with Z in the previous parameters' names. Similarly, the pointing selection an X and Z axes form the mean of the orbital parameters during the ScW are controled via the same parameters, except that `Ptg` is replaced by `OrbitPtg`.

Chapter 6

Masks modeling for pointing selections

Models and parameters used for masks definition in *spiselectscw* are taken from *gensskymodel*. Hereafter are some useful information taken from *gensskymodel* users manual.

In the following, sky coordinates can be equatorial, galactic or ecliptic system. Thus, *chi* (χ) will stand hereafter for the right ascension or the longitude and *psi* (ψ) for the declination or the latitude. All angles are expressed in degrees. There is no restriction on the validity range for χ but it is recommended to keep it in the $]-180:180]$ deg or in the $[0:360[$ deg ranges. Conversely, ψ is required to be in the $[-90:90]$ deg range.

6.1 Some words on spherical geometry

When dealing with models defined on a celestial frame, it is necessary to use spherical geometry and then define distances and angles on a sphere which are compatible with the same notions in Euclid's geometry (see fig. 6.1).

Distances The distance on the sky is defined as the angle between two points, i.e the length measured on a great circle from one point to the other, assuming the sphere has a radius of 1. With this “natural” definition, the distance is expressed in degrees between 0 and 180 and given by the formula :

$$\cos \rho = \sin \psi_1 \sin \psi_2 + \cos \psi_1 \cos \psi_2 \cos(\chi_2 - \chi_1),$$

where ρ is the angular distance and indexes 1 and 2 refer to the two points on the sky.

Angles To define the orientation of a celestial frame with respect to another, it is necessary to define rotation angles in a comparable manner as the trigonometric angles in a Cartesian

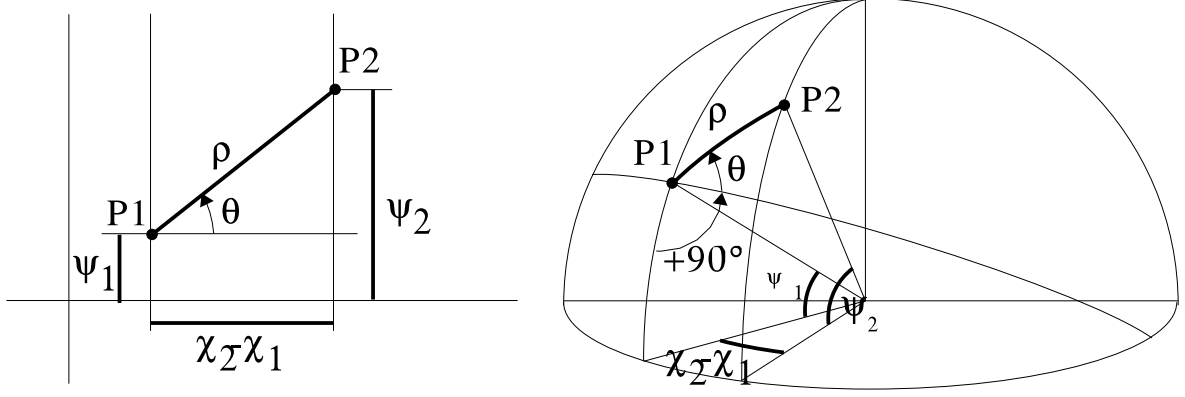


Figure 6.1: Distances and angles on a Cartesian (left) and a spherical (right) frames

referential. With the notations of fig. 6.1, we have :

$$\sin \theta = \begin{cases} \frac{\sin \psi_2 - \sin \psi_1 \cos \rho}{\cos \psi_1 \sin \rho} & \text{if } \psi_1 \neq 90^\circ [180] \\ -\frac{\sin \psi_1 \cos \psi_2 \cos(\chi_2 - \chi_1)}{\sin \rho} & \text{otherwise} \end{cases}$$

$$\cos \theta = \frac{\sin(\chi_2 - \chi_1) \cos \psi_2}{\sin \rho}$$

Note that since ρ is between 0 and 180 degrees, $\sin \rho \geq 0$, so, θ may be calculated using :

$$\tan \theta = \begin{cases} \frac{(\sin \psi_2 - \sin \psi_1 \cos \rho) / \cos \psi_1}{\sin(\chi_2 - \chi_1) \cos \psi_2} & \text{if } \psi_1 \neq 90^\circ [180] \\ -\frac{\sin \psi_1 \cos \psi_2 \cos(\chi_2 - \chi_1)}{\sin(\chi_2 - \chi_1) \cos \psi_2} & \text{otherwise} \end{cases},$$

where the quadrant of θ is determined by the signs of the numerator and denominator. This formula avoids the problem of a null distance.

6.2 Masks models description

Masks are used to define the portion of the sky selected for the X or Z pointing axes. Many masks models can be defined for each axis, their number for the axis $\langle A \rangle$ ($=X$ or Z) and data set $\langle DSet \rangle$ ($=Ptg$ or $OrbitPtg$) is given in `select_< DSet >< A >_masks_num`. When more than one mask is defined for an axis, they are logically combined according to the integer vector given in `(select_< DSet >< A >_masks_logeq`. In this vector (of size `select_< DSet >< A >_masks_num-1`), each component (0 for OR, 1 for AND) gives the relationship of a mask with the next one. For example, if one wants to evaluate `<mask#1>.OR.<mask#2>.AND.<mask#3>`, the logical vector would be : "0 1". Note that the priority rules apply, that is all AND statements will be evaluated first, and then the OR operations will be performed with the resulting blocks. For the example above, this means that `<mask#2>.AND.<mask#3>` will be evaluated first, and the result combined with OR to `<mask#1>`.

The global result of the masks can be revert (true becomes false, and vice versa) with the `select_< DSet >< A >_masks_globrev` set to 1.

The coordinates of the center of the masks `#m` for the axis `< A >` and data set `< DSet >` are given with `select_< DSet >< A >_masks_chi_list_#m` and `select_< DSet >< A >_masks_psi_list_#m` in the WCS defined by `select_< DSet >< A >_masks_coordsys_#m`. The result of the mask is evaluated in this WCS. The aspect rotation angle is `select_< DSet >< A >_masks_angle_list_#m`.

The mask name is defined by `select_< DSet >< A >_masks_name_#m` and must correspond to one of the following models (case sensitive). The result of any individual mask can be revert by setting `select_< DSet >< A >_masks_rev_#m` to 1. Each mask require a minimal number of parameters, given by `select_< DSet >< A >_masks_npar_#m` and `select_< DSet >< A >_masks_par_#m`, see the description of models for more information.

In the following descriptions, the reverse flag is assumed to be set to 0 (i.e. no reverse of the result).

True model : True

Name : True
Number of parameters : 0
Parameters description :
Model description : Returns 1 (true), whatever χ and ψ

This model returns always TRUE, basically used for test procedures or to switch on/off a mask.

False model : False

Name : False
Number of parameters : 0
Parameters description :
Model description : Returns 0 (false), whatever χ and ψ

This model returns always FALSE, basically used for test procedures or to switch on/off a mask.

Elliptical model : Ellipse

Name : Ellipse
Number of parameters : 4
Parameters description : χ coordinate of the center (χ_0) ; ψ coordinate of the center (ψ_0) ; semi-axis along χ (σ_χ) ; semi-axis along ψ (σ_ψ)
Model description : Returns 1 if $\rho \sqrt{\left(\frac{\cos \theta}{\sigma_\chi}\right)^2 + \left(\frac{\sin \theta}{\sigma_\psi}\right)^2} \leq 1$, 0 otherwise

Returns 1 (true) if χ and ψ are in the ellipse of center (χ_0, ψ_0) and semi-axes σ_χ, σ_ψ :

$$f(\chi, \psi) = \begin{cases} 1 & \text{if } \rho \sqrt{\left(\frac{\cos \theta}{\sigma_\chi}\right)^2 + \left(\frac{\sin \theta}{\sigma_\psi}\right)^2} \leq 1, \\ 0 & \text{otherwise.} \end{cases},$$

where the distance ρ and the angle θ are calculated using the formulas of § 6.1, using $(\chi_0; \psi_0)$ as the reference point (χ_1, ψ_1) in the formulas).

Note that to correctly treat the equality case with numerical approximations, the boundaries are shifted by 10^{-6} degrees outward the theoretical “good” interval.

Double elliptical model : Ellipse2

Name	: Ellipse2
Number of parameters	: 6
Parameters description	: χ coordinate of the center (χ_0) ; ψ coordinate of the center (ψ_0) ; inner semi-axis along χ (σ_χ^0) ; inner semi-axis along ψ (σ_ψ^0) ; outer semi-axis along χ (σ_χ^1) ; outer semi-axis along ψ (σ_ψ^1)
Model description	: returns 1 if $\rho \sqrt{\left(\frac{\cos \theta}{\sigma_\chi^0}\right)^2 + \left(\frac{\sin \theta}{\sigma_\psi^0}\right)^2} \geq 1$ and $\rho \sqrt{\left(\frac{\cos \theta}{\sigma_\chi^1}\right)^2 + \left(\frac{\sin \theta}{\sigma_\psi^1}\right)^2} \leq 1$, : returns 0 otherwise

Returns 1 if (χ, ψ) is in an region delimited by two ellipses of semi-axes σ_χ^0 and σ_ψ^0 for the inner one and σ_χ^1 and σ_ψ^1 for the outer one. Each of them is centered on $(0,0)$:

$$f(\chi, \psi) = \begin{cases} 1 & \text{if } \rho \sqrt{\left(\frac{\cos \theta}{\sigma_\chi^0}\right)^2 + \left(\frac{\sin \theta}{\sigma_\psi^0}\right)^2} \geq 1 \text{ and } \rho \sqrt{\left(\frac{\cos \theta}{\sigma_\chi^1}\right)^2 + \left(\frac{\sin \theta}{\sigma_\psi^1}\right)^2} \leq 1, \\ 0 & \text{otherwise.} \end{cases},$$

where the distance ρ and the angle θ are calculated using the formulas of § 6.1, using $(\chi_0; \psi_0)$ as the reference point (χ_1, ψ_1) in the formulas).

Note that to correctly treat the equality case with numerical approximations, the boundaries are shifted by 10^{-6} degrees outward the theoretical “good” interval.

Positive latitude model : LatSup

Name	: LatSup
Number of parameters	: 1
Parameters description	: Latitude threshold (ψ_0)
Model description	: Returns 1 if $\psi \geq \psi_0$, 0 otherwise

Returns 1 (true) if the latitude ψ is greater or equal to ψ_0 :

$$f(\chi, \psi) = \begin{cases} 1 & \text{if } \psi \geq \psi_0, \\ 0 & \text{otherwise.} \end{cases},$$

Note that to correctly treat the equality case with numerical approximations, the boundaries are shifted by 10^{-6} degrees outwards the theoretical “good” interval.

Negative latitude model : LatInf

Name : LatInf
Number of parameters : 1
Parameters description : Latitude threshold (ψ_0)
Model description : Returns 1 if $\psi \leq \psi_0$, 0 otherwise

Returns 1 (true) if the latitude ψ is smaller or equal to ψ_0 :

$$f(\chi, \psi) = \begin{cases} 1 & \text{if } \psi \leq \psi_0, \\ 0 & \text{otherwise.} \end{cases},$$

Note that to correctly treat the equality case with numerical approximations, the boundaries are shifted by 10^{-6} degrees outward the theoretical “good” interval.

Positive longitude model : LonSup

Name : LonSup
Number of parameters : 1
Parameters description : Longitude threshold (χ_0)
Model description : Returns 1 if $\chi \geq \chi_0$, 0 otherwise

Returns 1 (true) if the longitude χ is greater or equal to χ_0 :

$$f(\chi, \psi) = \begin{cases} 1 & \text{if } \chi \geq \chi_0, \\ 0 & \text{otherwise.} \end{cases},$$

Note that to correctly treat the equality case with numerical approximations, the boundaries are shifted by 10^{-6} degrees outward the theoretical “good” interval.

Negative longitude model : LonInf

Name : LonInf
Number of parameters : 1
Parameters description : Longitude threshold (χ_0)
Model description : Returns 1 if $\chi \leq \chi_0$, 0 otherwise

Returns 1 (true) if the longitude χ is smaller or equal to χ_0 :

$$f(\chi, \psi) = \begin{cases} 1 & \text{if } \chi \leq \chi_0, \\ 0 & \text{otherwise.} \end{cases},$$

Note that to correctly treat the equality case with numerical approximations, the boundaries are shifted by 10^{-6} degrees outward the theoretical “good” interval.

Rectangular model : Rect

Name : Rect

Number of parameters : 4

Parameters description : χ coordinate of the center (χ_0) ; ψ coordinate of the center (ψ_0) ; half size of the region along χ (w_χ) ; half size of the region along ψ (w_ψ)

Model description : Returns 1 if $|\chi - \chi_0| \leq w_\chi$ and $|\psi - \psi_0| \leq w_\psi$

Returns 1 (true) if the point $(\chi; \psi)$ is in a rectangular region :

$$f(\chi, \psi) = \begin{cases} 1 & \text{if } |\chi - \chi_0| \leq w_\chi \text{ and } |\psi - \psi_0| \leq w_\psi, \\ 0 & \text{otherwise.} \end{cases},$$

Note that to correctly treat the equality case with numerical approximations, the boundaries are shifted by 10^{-6} degrees outward the theoretical “good” interval.

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